Mth101 grand quiz spring 2021

Calculus (Virtual University of Pakistan)

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Grand Quiz Spring 2021

Subject Code MTH101 lecture 1 to 22

Solved by Rizwan Qadeer

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I'm providing 100% correct quiz solution. You can visit my YouTube channel and get more information about all other subjects' quizzes and final year project (CS619).

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MTH101:Grand Quiz

Question # 1 of 30 (Start time: 09:01:09 AM, 21 June 2021)

Is the function f(x) = tan (x) continuous at
x = pi ? If not, why?

Select the correct option

- (●) f is continuous at x = pi .
- ( ) f (pi) is not defined.



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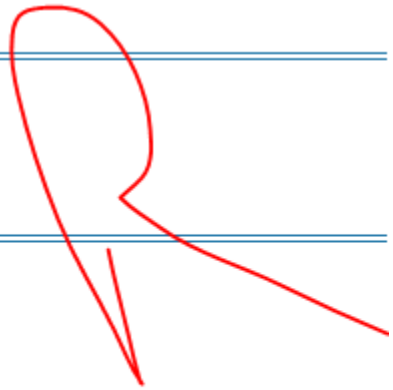
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Question # 2 of 30 (Start time: 09:01:29 AM, 21 June 2021)

For the curve: f(t) = -2t+3, the average rate of change of 'f(t)' over the interval [-1,0] is _____.

Select the correct option

- ○ 1
- ○ 2
- ○ -1
- ● -2



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Quiz Start Time: 09:01 AM

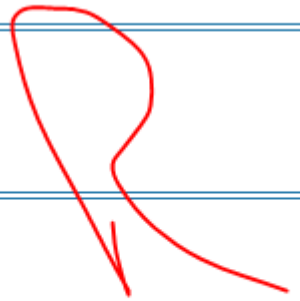
Question # 3 of 30 (Start time: 09:01:48 AM, 21 June 2021)

Let f be defined on an interval, and x_1, x_2 are the points in that interval. Then f is increasing on the interval if $f(x_1) < f(x_2)$ whenever

Select the correct option

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- $x_1 > x_2$
- $x_1 \geqslant x_2$
- $x_1 < x_2$
- $x_1 = x_2$



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Quiz Start Time:

Question # 4 of 30 (Start time: 09:02:05 AM, 21 June 2021)

$If\ g(X) = 3x^2\ then\ g'(2)\ will\ be\ .....\ ?$

Select the correct option

- -12
- 18
- 12
- 15

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Quiz Start Time: 09:01 AM.

Question # 5 of 30 (Start time: 09:02:20 AM, 21 June 2021)

$$Let \;\; f(x) = x + 5 \;\; and \;\; g(x) = x^2 - 3 \;\; then \;\; f(g(x)) = ?$$

Select the correct option

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○	$x^2 - 2$
○	$2x^2 + 2$
●	$x^2 + 2$
○	$x^2 + 1$



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Question # 6 of 30 (Start time: 09:02:36 AM, 21 June 2021)

If y=2x then instantaneous rate of change of 'y' w.r.t 'x' at 'x =-2' is .....

Select the correct option

○	-1
○	3
○	1
●	2

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Quiz Start Time:

Question # 7 of 30 (Start time: 09:02:54 AM, 21 June 2021)

$\lim\ [f(x)]^n =$

Select the correct option

- $[\lim f(x)]$
- $[f(x)]^n$
- $[\lim f(x)]^n$
- $n[\lim f(x)]$

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Quiz Start Time: 0

Question # 8 of 30 (Start time: 09:03:09 AM, 21 June 2021)

$\log_b M^p =$ _____.

Select the correct option

Rel

- $-\log_b M$
- $p \log_b M$
- $\log_b \frac{p}{M}$
- $M \log_b p$

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Question # 9 of 30 (Start time: 09:03:34 AM, 21 June 2021)

If the function f and g are continuous at c, then
f o g is ..... at c.

Select the correct option

- ○ Discontinuous
- ● Continuous



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Question # 10 of 30 (Start time: 09:04:07 AM, 21 June 2021)

If

$$f(x) = 4x^2 + 2x$$

, then which of the following is true about it.

Select the correct option

●	Its derivative with respect to x is 8x + 2.
○	None of these.
○	Its derivative with respect to x is 8x.
○	Its derivative with respect to x is 2x.

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Quiz Start Time: 09:01

Question # 11 of 30 (Start time: 09:04:26 AM, 21 June 2021)

If $f(x) = \cos(2x)$ *then its derivative by chain rule is* . . . . . .

Select the correct option

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- $\sin(x).2$
- $-\sin(2x).2$
- $-\sin(x).2$
- $\sin(2x).2$

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Quiz Start Time: 09:01

Question # 12 of 30 (Start time: 09:04:45 AM, 21 June 2021)

$If\ f(x) = x^5\ and\ g(x) = x^2,\ then\ the derivative\ of\ f(x) \times g(x)\ is$

Select the correct option

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○	$3x^2$
●	None of these
○	$6x^3$
○	$7x^2$

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Quiz Start Time: 09:01 AM, 2

Question # 13 of 30 (Start time: 09:05:06 AM, 21 June 2021)

T

$$\frac{d}{dx}(\ln u) = _____ .$$

Select the correct option

Reload Math

- ● $\frac{1}{u} \cdot \frac{du}{dx}$
- ○ $\frac{1}{u}$
- ○ $\frac{du}{dx}$
- ○ $x \ln x - x + C$

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Question # 14 of 30 (Start time: 09:05:24 AM, 21 June 2021)

If the graph of a function oscillates between 1 and -1 then its limit is

Select the correct option

- 0
- Does not exist
- 1
- -1

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Question # 15 of 30 (Start time: 09:06:05 AM, 21 June 2021)

If f(x) = sin(x) and g(x) = sin(2x), then the derivative of f(x) + g(x) is _____

Select the correct option

○	None of these.
●	cos(x) + 2cos(2x).
○	cos(x) + 2sin(x).
○	cos(x) + sin(2x).



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Quiz Start Time: 09:01 AM.

Question # 16 of 30 (Start time: 09:06:33 AM, 21 June 2021)

$$If\ f(x) = sinx\ and\ g(x) = x^2,\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

Reload Mat

○	None of these.
○	$[x^2cos(x) - 2xsin(x)]/x^2$
●	$[x^2cos(x) - 2xsin(x)]/x^4$
○	$[x^2cos(x) + 2xsin(x)]/x^4$

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Question # 17 of 30 (Start time: 09:06:48 AM, 21 June 2021)

A horizontal line has slpoe _____.

Select the correct option

- 2
- 3
- 1
- 0



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Quiz Start Time: 09:01 AM.

Question # 18 of 30 (Start time: 09:07:48 AM, 21 June 2021)

The vertical asymptote of the function $f(x) = \dfrac{3x^2 + 1}{2 - x}$ is

Select the correct option

Reload Mat

- (●) 2
- ( ) -2
- ( ) ∞
- ( ) 0

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Quiz Start Time: 09:01 AM, 21 Jun

Question # 20 of 30 (Start time: 09:08:31 AM, 21 June 2021)

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The function f is said to be _____ on (a, b), if $f''(x) > 0$ on that interval.

Select the correct option

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- None of these
- Concave up
- Constant
- Concave down



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Question # 21 of 30 (Start time: 09:09:33 AM, 21 June 2021)

What is the derivative of

$$7x^3$$

Select the correct option

- ● $21x^2$
- ○ $21x^3$
- ○ None of these
- ○ $7x^4$



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Question # 22 of 30 (Start time: 09:09:47 AM, 21 June 2021)

The equation of the tangent line passing through the point (-3,2) having the slope 4 is

Select the correct option

- y+2=-4(x+3)
- y+2=4(x+3)
- y-2=4(x-3)
- y-2=4(x+3)



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Quiz Start Time: 09:01 AM,

Question # 24 of 30 (Start time: 09:11:12 AM, 21 June 2021)

$If\ f(x) = 2x + 2\ and\ g(x) = x^2 + 3x,\ then\ the\ derivative\ of\ f(x) + g(x)\ is$

Select the correct option

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○	2 + 5x
○	3 + 5x
○	None of these
●	5 + 2x

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Quiz Start Time: 09:01 AM, 21 Ju

Question # 25 of 30 (Start time: 09:11:33 AM, 21 June 2021)

Total

What is the derivative of $2x^2 \sin(x^5)$?

Select the correct option

Reload Math Equ

○	$2x^2 \cos(x^5)$
○	$-10x^2 \cos(x^5)$
○	$10x^2 \cos(x^5)$
●	None of these

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Quiz Start Time: 09:0

Question # 26 of 30 (Start time: 09:11:56 AM, 21 June 2021)

What is the derivative of 3cot (2x)

Select the correct option

Reloa

- ● $-6cosec^2(2x)$
- ○ None of these
- ○ $6cosec^2(2x)$
- ○ $-3cosec^2(2x)$

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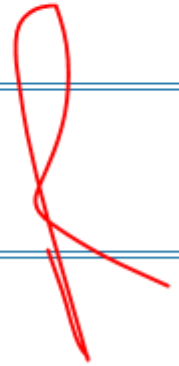
Quiz Start Time:

Question # 27 of 30 (Start time: 09:12:11 AM, 21 June 2021)

If $y = x^2 + 9x$, then $\frac{dy}{dx} =$ _____.

Select the correct option

- (●) $2x + 9$
- ( ) 9
- ( ) $2x$
- ( ) $2x + 9x$



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Question # 28 of 30 (Start time: 09:12:27 AM, 21 June 2021)

The slope of tangent lines for the graph of constant function is

Select the correct option

●	0
○	Undefined
○	1
○	-1



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Quiz Start Time: (

Question # 29 of 30 (Start time: 09:13:11 AM, 21 June 2021)

What is the derivative of $\sin(x^2)$?

Select the correct option

Re

○	None of these
○	$2 \cos x^2$
●	$2x \cos x^2$
○	$2 \cos x$

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Quiz Start Time: 09:01 AM, 21 Ju

Question # 30 of 30 (Start time: 09:13:27 AM, 21 June 2021)

Total

$$If\ f(x) = x^5 + sin2x + tan2x + cos3x,\ then\ derivative\ of\ f(x)\ will\ be$$

Select the correct option

Reload Math Equ

- $5x^4 + 2cos(2x) + 2sec^2(2x) - 3sin(3x).$
- $5x^4 + 2cos(2x) + 2sec^2(2x) + 3sin(3x).$
- None of these.
- $5x^4 + 2cos(2x) + sec^2(2x) - 3sin(3x)$

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